



Biology 1 Performance Targets

for the
South Carolina College- and Career-Ready Science Standards 2021

For use 2026-2027

July 2026

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Purpose and Use

Science is a way of understanding the physical universe using observation and experimentation to explain natural phenomena. Science also refers to an organized body of knowledge that includes core ideas to the disciplines and common themes that bridge the disciplines. As science educators we must take a 3-dimensional approach in facilitating student learning. By addressing content, science and engineering practices and crosscutting concepts, students can have relevant and evidence-based instruction that can help solve current and future problems.

This document is intended as a guide for discerning and describing features of students and their work who have met the stated Performance Expectation (PE). This document is not intended to be read from cover to cover, but to be used, when needed, to support teacher professional learning and curriculum decisions. This is not intended for student use, and thus is not written in student-friendly language. This is not a curriculum or a means to limit instruction in the classroom. Although each PE states a dedicated Science and Engineering Practice (SEP) and Crosscutting Concept (CCC), students will need to use the whole range of SEPs and CCCs to achieve success by the end of instruction.

Three-dimensional science learning requires discipline specific communication skills. This means that effective science learning occurs when students are expected to speak, listen, read, and write in ways that are appropriate to science. With each Performance Target, there are question/sentence stems and terminology to support student discourse about phenomena to help teachers facilitate the acquisition of science discourse. Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding. The terms and stems in this section are intended to provide a baseline for teachers, neither list is exhaustive nor complete.

In addition to the doing (SEP), thinking (CCC), and learning of science knowledge (Disciplinary Core Ideas) outlined here, students will also require a working knowledge of grade-level appropriate tools and techniques of science. Students should know and recognize how scientists and engineers use these tools and techniques, not just identify them. Students should be able to use these tools to gather data, describe how these tools gather data, and/or interpret data sampled from them.

Updates

July 2026

- Addition of Appendix A: Biology 1 Vertical Articulation of Disciplinary Core Idea Instructional Focus Statements
- Additional terminology:
 - B-LS1-5
 - granum
 - B-LS2-7
 - relative abundance
 - species richness
- Adjusted formatting and grammar.

July 2025

- All Performance Expectation statements have been reformatted to call out each of the dimensions as follows:
 - Science and Engineering Practice – **bold**
 - Crosscutting Concept – *italicize*
 - Disciplinary Core Idea – regular
- The watermark from previous versions of this resource has been replaced with the wording “For use 2025-2026” on the title page and in the footer. This change was made to improve accessibility of this resource.
- Because scientific notation is no longer an expectation of the math standards, the following PEs have a new statement related to the use and application of scientific notation.
 - B-LS1-1
 - B-LS1-6
 - B-LS2-1
 - B-LS4-1
- B-LS1-6
 - How these biomolecules (limited to: carbohydrates, lipids, nucleic acids, and proteins) function to support organisms.
 - The energy released in cellular respiration can be used to drive chemical reactions between sugars and other substances to produce amino acids and other biomolecules, which function to support organisms.
- Additional terminology
 - B-LS1-1
 - peptide
 - B-LS1-6
 - peptide

- B-LS2-1
 - autotroph
 - consumer
 - heterotroph
 - producer
- B-LS2-5
 - zooplankton
- B-LS4-4
 - allele frequency
- B-LS4-5
 - allele frequency

June 2024

- Updated watermark to 2024-2025.
- Adjusted formatting and grammar.

LS1 – From Molecules to Organisms: Structures and Processes

B-LS1-1. Construct an explanation based on evidence for how the structure of DNA determines *the structure of proteins, which carry out the essential functions* of life through systems of specialized cells.

State Assessment Boundary: Assessment does not include identification of specific cell or tissue types, whole body systems, specific protein structures and functions, or the biochemistry of protein synthesis.

Science and Engineering Practices	Disciplinary Core Ideas	Crosscutting Concepts
Constructing Explanations and Designing Solutions Constructing explanations and designing solutions in 9-12 builds on K-8 experiences and progresses to explanations and designs that are supported by multiple and independent student-generated sources of evidence consistent with scientific ideas, principles, and theories. Construct an explanation based on valid and reliable evidence obtained from a variety of sources (including [models, peer review, simulations, theories, students' own investigations]) and the assumption that theories and laws that describe the natural world operate today as they did in the past and will continue to do so in the future.	LS1.A: Structure and Function Systems of specialized cells within organisms help them perform the essential functions of life. All cells contain genetic information, in the form of DNA. Genes are specific regions within the extremely large DNA molecules that form the chromosomes. Genes contain the instructions that code for the formation of molecules called proteins, which carry out most of the work of cells to perform the essential functions of life. Proteins provide structural components, serve as signaling devices, regulate cell activities, and determine the performance of cells through their enzymatic actions. LS3.A: Inheritance of Traits The sequence of nucleotides spells out the information in a gene. DNA controls the expression of proteins by being transcribed into a “messenger” RNA, which is translated in turn by the cellular machinery into a protein.	Structure and Function Investigating or designing new systems or structures requires a detailed examination of the properties of different materials, the structures of different components, and connections of components to reveal its function.

*Because DNA molecules contain billions of base pairs, students may need to understand and apply the conventions of scientific notation when working with quantities of measurement and their calculations.

Observable features of student performance by the end of the course:

1. Articulating the explanations of phenomena

- a. Students articulate a statement describing/explaining relationships between the structure of DNA and the structure and function of resulting proteins, including:
 - i. regions of DNA (limited to genes) determine the structure of proteins, and
 - ii. proteins carry out the essential functions of life.

2. Evidence

- a. Students identify and describe the evidence to construct the explanation, including:
 - i. all cells contain DNA,
 - ii. DNA contains regions called genes,
 - iii. the sequence of genes contains instructions that code for proteins,
 - iv. a single gene codes for a protein,
 - v. through the process of transcription, a DNA sequence is transcribed into a mRNA sequence,
 - vi. through the process of translation, a mRNA sequence is translated into an amino acid sequence (limited to a protein) at the ribosome,
 - vii. proteins carry out the essential functions of life, and
 - viii. in complex organisms, groups of specialized cells (limited to tissues) use proteins to carry out functions that are essential to the organism.
- b. Students use multiple valid and reliable sources of evidence, including student experiments.

3. Reasoning

- a. Students use the following chain of reasoning to connect the evidence and support, refute, or revise an explanation, including:
 - i. because all cells contain DNA, all cells contain genes that can code for the formation of proteins,
 - ii. gene sequence affects protein function,
 - iii. proper functioning of many proteins is necessary for the proper functioning of cells,
 - iv. as a result of differentiation, different genes are activated in different cells that share the same genetic code in complex organisms, and
 - v. systems of specialized cells (limited to tissues) share similar structures and functions, these are carried out by proteins.

B-LS1-1 Academic Language

Question/Sentence Stems

- The _____ structures help _____ to function because _____.
- The _____ structures are present in _____ and are related to the function _____.

Terminology to Support Student Discourse about Phenomena

*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- | | |
|--|--------------------------|
| • Adenine (A) | • nuclear membrane |
| • amino acid | • nucleic acid |
| • anticodon | • nucleotide |
| • base | • nucleus |
| • cell | • nucleus |
| • chromosome | • peptide |
| • codon | • peptide bond |
| • cytoplasm | • phosphate |
| • Cytosine (C) | • polypeptide |
| • deoxyribose | • protein synthesis |
| • differentiation | • ribose |
| • deoxyribonucleic acid (DNA) | • ribosome |
| • double helix | • ribonucleic acid (RNA) |
| • endoplasmic reticulum (smooth and rough) | • ribosomal RNA (rRNA) |
| • enzyme | • start codon |
| • gene | • stop codon |
| • Golgi apparatus | • Thymine (T) |
| • Guanine (G) | • transcription |
| • hydrogen bond | • translation |
| • messenger RNA (mRNA) | • transfer RNA (tRNA) |
| • mutation | • Uracil (U) |
| | • vesicle |

B-LS1-4. Use a model to illustrate the role of cellular division (mitosis) and differentiation in producing and maintaining complex organisms.

Clarification Statement: Emphasis is on normal cell division as well as instances in which cell division is uncontrolled (e.g., cancer).

State Assessment Boundary: Assessment does not include specific gene control mechanisms or rote memorization of the steps of mitosis.

Science and Engineering Practices	Disciplinary Core Ideas	Crosscutting Concepts
Developing and Using Models Modeling in 9-12 builds on K-8 experiences and progresses to using, synthesizing, and developing models to predict and show relationships among variables between systems and their components in the natural and designed worlds. Use a model based on evidence to illustrate and/or predict the relationships between systems or between components of a system.	LS1.B: Growth and Development of Organisms In multicellular organisms, individual cells grow and then divide via a process called mitosis [sic], thereby allowing the organism to grow (and repair). The organism begins as a single cell (fertilized egg) that divides successively to produce many cells, with each parent cell passing identical genetic material (two variants of each chromosome pair) to both daughter cells. Cellular division and differentiation produce and maintain a complex organism, composed of systems of tissues and organs that work together to meet the needs of the whole organism.	Systems and System Models Models (e.g., physical, mathematical, computer models) can be used to simulate systems and interactions—including energy, matter, and information flows—within and between systems at different scales.

*Mitosis is the division of the nucleus; cytokinesis is division of the cytoplasm.

Observable features of student performance by the end of the course:

1. Components of the model

- a. Students develop/use a model (conceptual, graphical, physical, etc.) and identify the relevant components to illustrate the role of mitosis and differentiation in producing and maintaining complex organisms, including:
 - i. genetic material containing two variants of each chromosome pair (one for each parent),
 - ii. parent and daughter cells,
 - iii. a multicellular organism as a collection of differentiated cells, and/or

- iv. phases of the cell cycle, including:
 - 1. Gap 1 (G1 phase),
 - 2. Synthesis (S phase),
 - 3. Gap 2 (G2 phase),
 - 4. Mitosis (M phase),
 - 5. cytokinesis, and
 - 6. checkpoints.

2. Relationships

- a. Students develop/use a model to describe the relationships between components, including:
 - i. daughter cells receive identical genetic information from a parent cell (including a fertilized egg),
 - ii. the cell cycle is the process through which cells replicate, producing two genetically identical daughter cells,
 - iii. mitotic division produces two genetically identical daughter cells from one parent cell (for example: describe the sequence of events during mitosis, etc.),
 - iv. when cells bypass the checkpoints of the cell cycle, uncontrolled cell division (limited to cancer) may occur, and
 - v. differences between cell types within a multicellular organism are due to gene expression and not genetic differences between the cells.

3. Connections

- a. Students develop/use a model to demonstrate that mitotic division results in more cells that:
 - i. allow growth of the organism,
 - ii. differentiation, and/or
 - iii. replace damaged/dead cells to maintain a complex organism.
- b. Students develop/use a model to predict what will happen during the cell cycle/mitosis under different conditions (for example: did not complete DNA replication, cell bypasses a checkpoint, etc.).
- c. Students identify limits of the model (for example: accuracy of the model and the actual process of cellular division, etc.).

B-LS1-4 Academic Language

Question/Sentence Stems

- The key components of the system are_____.
- In the system, _____ and _____ are shown in the model.
- In the system, _____ and _____ work together to_____.
- In the system, _____ and _____ interact in _____ way.

Terminology to Support Student Discourse about Phenomena

*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- anaphase
- cancer
- carcinogen
- cell cycle
- cell plate
- centromere
- centrosome
- checkpoint
- chromosome
- cleavage furrow
- cytokinesis
- cytoplasm
- daughter cell
- differential gene expression
- differentiation
- diploid
- deoxyribonucleic acid (DNA)
- egg cell
- embryo
- fertilize
- gap 1 (G_1)
- gap 2 (G_2)
- gene
- genome
- growth
- haploid
- interphase
- maintenance
- metaphase
- mitosis
- mitotic phase (M)
- multicellular
- nucleus
- parent cell
- prophase
- protein
- replication
- sister chromatid
- somatic cell
- sperm cell
- spindle fibers
- stem cell
- synthesis phase (S)
- telophase
- tumor (benign and malignant)
- unicellular
- zygote

B-LS1-5. Use a model to illustrate how photosynthesis *transforms light energy into stored chemical energy.*

Clarification Statement: Emphasis is on explaining inputs and outputs of matter and the transfer and transformation of energy in photosynthesis by plants and other photosynthesizing organisms. Examples of models could include diagrams, chemical equations, and conceptual models.

State Assessment Boundary: Assessment does not include specific biochemical steps.

Science and Engineering Practices	Disciplinary Core Ideas	Crosscutting Concepts
Developing and Using Models Modeling in 9-12 builds on K-8 experiences and progresses to using, synthesizing, and developing models to predict and show relationships among variables between systems and their components in the natural and designed worlds. Use a model based on evidence to illustrate the relationships between systems or between components of a system.	LS1.C: Organization for Matter and Energy Flow in Organisms The process of photosynthesis converts light energy to stored chemical energy by converting carbon dioxide plus water into sugars plus released oxygen.	Energy and Matter Changes of energy and matter in a system can be described in terms of energy and matter flows into, out of, and within that system.

Observable features of student performance by the end of the course:

1. Components of the model

- a. Students develop/use a model (conceptual, graphical, physical, etc.) and identify the relevant components to illustrate the function of photosynthesis including:
 - i. chloroplast,
 - ii. solar energy,
 - iii. chemical energy transfer between subsystems (light-dependent and light-independent reactions):
 1. ATP/ADP and
 2. NADPH/NADP⁺, and
 - iv. matter:
 1. CO₂,
 2. H₂O,
 3. C₆H₁₂O₆, and
 4. O₂.

2. Relationships

- a. Students develop/use a model to describe relationships between the components, including:
 - i. inputs,
 - ii. outputs, and
 - iii. energy transfer between subsystems.

3. Connections

- a. Students develop/use a model to demonstrate:
 - i. The transfer of matter and flow of energy between an organism and its environment during photosynthesis.
 - ii. The transfer of matter and flow of energy within the cell of a photosynthetic organism.
 - iii. Photosynthesis results in the storage of energy in the difference between the energies of the chemical bonds of the inputs and the outputs.

B-LS1-5 Academic Language

Question/Sentence Stems

- _____ happens to matter as it moves within the system.
- In this system, energy is entering by _____, doing _____ in the system, and leaving the system by _____.
- The energy for _____ is from _____.
- The flow of energy causes _____ to occur in the system.
- The energy is entering the system by _____.
- In the system, the cycling of matter _____.
- The matter in the system enters from _____.
- When the matter leaves the system, it goes _____.

Terminology to Support Student Discourse about Phenomena

*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- | | | |
|--------------------------------|---------------------|------------------|
| • adenosine diphosphate (ADP) | • chloroplasts | • output |
| • adenosine triphosphate (ATP) | • energy flow | • oxygen |
| • Calvin cycle | • glucose | • photosynthesis |
| • carbon dioxide | • granum | • product |
| • carbon fixation | • input | • reactant |
| • cellular respiration | • light-dependent | • simple sugar |
| • chemical reaction | • light-independent | • solar energy |
| • chlorophyll | • matter cycle | • stroma |
| | • mitochondrion | • thylakoid |
| | • NADP+ | • water |
| | • NADPH | |

B-LS1-6. Construct and revise an explanation based on evidence for how carbon, hydrogen, and oxygen from sugar molecules *may combine with other elements to form* amino acids and other large carbon-based molecules necessary for essential life processes.

Clarification Statement: Emphasis is on using evidence from models and simulations to support explanations of how the products of photosynthesis can be used to form the molecules of life.

State Assessment Boundary: Assessment does not include the details of the specific chemical reactions or molecular identification of macromolecules.

Science and Engineering Practices	Disciplinary Core Ideas	Crosscutting Concepts
Constructing Explanations and Designing Solutions Constructing explanations and designing solutions in 9-12 builds on K-8 experiences and progresses to explanations and designs that are supported by multiple and independent student-generated sources of evidence consistent with scientific ideas, principles, and theories. Construct and revise an explanation based on valid and reliable evidence obtained from a variety of sources (including [models, peer review, simulations, theories, students' own investigations]) and the assumption that theories and laws that describe the natural world operate today as they did in the past and will continue to do so in the future.	LS1.C: Organization for Matter and Energy Flow in Organisms The sugar molecules thus formed contain carbon, hydrogen, and oxygen: the hydrocarbon backbones are used to make amino acids and other carbon-based molecules that can be assembled into larger molecules (such as proteins or DNA), used for essential life functions. As matter and energy flow through organizational levels of living systems, chemical elements are recombined to form different products.	Energy and Matter Changes of energy and matter in a system can be described in terms of energy and matter flows into, out of, and within that system.

*Because biomolecules can be quite small, students may need to understand and apply the conventions of scientific notation when working with quantities of measurement and their calculations.

Observable features of student performance by the end of the course:

1. Articulating the explanation of phenomena

- a. Students articulate a statement describing/explaining the relationship between the products of photosynthesis and other biomolecules necessary for organism function, including:
 - i. The relationship between the carbon, hydrogen, and oxygen atoms from sugar molecules formed in or ingested by an organism and those same atoms found in amino acids and other macromolecules.
 - ii. Biomolecules and amino acids can be a result of chemical reactions between photosynthesis products (or component atoms) and other atoms.
 - iii. That the energy for building these macromolecules comes from ATP produced during the process of cellular respiration.

2. Evidence

- a. Students identify and describe the evidence to construct the explanation, including:
 - i. All organisms take in matter and rearrange the atoms through chemical reactions.
 - ii. Sugar molecules (for example: glucose, etc.) are the energy storage molecules produced by photosynthetic organisms.
 - iii. Sugar molecules (for example: glucose, etc.) are composed of carbon, oxygen, and hydrogen atoms.
 - iv. Amino acids and other biomolecules (for example: carbohydrates, lipids, nucleic acids) are mostly composed of carbon, hydrogen, and oxygen atoms.
 - v. How these biomolecules (limited to: carbohydrates, lipids, nucleic acids, and proteins) function to support organisms.
 - vi. Chemical reactions can create products that are more complex than the reactants.
 - vii. Chemical reactions involve changes in the energies of the molecules involved in the reaction.
- b. Students use multiple valid and reliable sources of evidence, including student experiments.

3. Reasoning

- a. Students use the following chain of reasoning to connect the evidence and support, refute, or revise an explanation, including:
 - i. The atoms in sugar molecules can provide most of the atoms that comprise amino acids and other biomolecules.
 - ii. The energy released in cellular respiration can be used to drive chemical reactions between sugars and other substances to produce amino acids and other biomolecules, which function to support organisms.
 - iii. The matter flows in cellular processes are the result of the rearrangements of primarily atoms in sugar molecules produced through photosynthesis because those are the molecules whose reactions release the energy needed for cell processes.

B-LS1-6 Academic Language

Question/Sentence Stems

- The matter in the system enters from _____.
- When the matter leaves the system, it goes _____.
- The flow of energy causes _____ to occur in the system.
- In the system, the cycling of matter _____.

Terminology to Support Student Discourse about Phenomena

*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- | | | |
|-------------------------------------|-----------------|----------------------------|
| • adenosine triphosphate (ATP) | • enzyme | • organic molecules |
| • amino acid | • fatty acid | • oxygen (O ₂) |
| • bond | • glucose | • peptide |
| • carbohydrate | • hydrocarbon | • phosphorus (P) |
| • carbon (C) | • hydrogen (H) | • photosynthesis |
| • carbon dioxide (CO ₂) | • lipid | • product |
| • cellular respiration | • macromolecule | • protein |
| • cellulose | • matter | • reactant |
| • chemical reaction | • molecule | • rearrangement |
| • energy | • nitrogen (N) | • starch |
| | • nucleic acid | • sugar |
| | • nucleotide | • sulfur (S) |
| | • organic | |

B-LS1-7. Use a model to illustrate that cellular respiration is a chemical process whereby the bonds of food molecules are broken and the bonds in new compounds are formed, resulting in *a net transfer of energy*.

Clarification Statement: Emphasis is on the conceptual understanding of the inputs and outputs of the processes of aerobic and anaerobic cellular respiration.

State Assessment Boundary: Assessment should not include identification of the steps or specific processes involved in cellular respiration nor specific types of fermentation. Assessment should be limited to comparing efficiency of aerobic and anaerobic cellular respiration.

Science and Engineering Practices	Disciplinary Core Ideas	Crosscutting Concepts
Developing and Using Models Modeling in 9-12 builds on K-8 experiences and progresses to using, synthesizing, and developing models to predict and show relationships among variables between systems and their components in the natural and designed worlds. Use a model based on evidence to illustrate the relationships between systems or between components of a system.	LS1.C: Organization for Matter and Energy Flow in Organisms As matter and energy flow through different organizational levels of living systems, chemical elements are recombined in different ways to form different products. As a result of these chemical reactions, energy is transferred from one system of interacting molecules to another. Cellular respiration is a chemical process in which the bonds of food molecules and oxygen molecules are broken, and new compounds are formed that can transport energy to muscles. Cellular respiration also releases the energy needed to maintain body temperature despite ongoing energy transfer to the surrounding environment. Anaerobic cellular respiration follows a different and less efficient chemical pathway to provide energy in cells.	Energy and Matter Energy cannot be created or destroyed—it only moves between one place and another place, between objects and/or fields, or between systems.

Observable features of student performance by the end of the course:

1. Components of the model

- a. Students develop/use a model (conceptual, graphical, physical, etc.) to identify the relevant components to illustrate cellular respiration, including:
 - i. mitochondrion,
 - ii. cytoplasm,
 - iii. matter:
 1. glucose ($C_6H_{12}O_6$),
 2. CO_2 ,
 3. H_2O , and
 4. O_2 ,
 - iv. transfer of matter and energy between the following molecular systems:
 1. glycolysis,
 2. Kreb's cycle,
 3. Electron Transport Chain, and
 4. fermentation, and
 - v. energy transfer and release:
 1. ATP/ADP and
 2. NADH/NAD⁺.

2. Relationships

- a. Students develop/use a model to describe the relationships between the components, including:
 - i. inputs,
 - ii. outputs,
 - iii. energy transfer and conservation of matter between subsystems, and
 - iv. efficiency.

3. Connections

- a. Students develop/use a model to demonstrate:
 - i. The process of cellular respiration releases energy as the matter is rearranged, existing chemical bonds are broken, and new chemical bonds are formed, but matter and energy are neither created nor destroyed.
 - ii. The process of cellular respiration transfers energy to the cell to sustain life's processes.
 - iii. When oxygen is not available, cells will use the process of fermentation to maintain ATP production to continue cell work (fermentation/anaerobic respiration), with a marked decrease in ATP production efficiency as compared to aerobic respiration.

B-LS1-7 Academic Language

Question/Sentence Stems

- _____ happens to matter as it moves within the system.
- In this system, energy is entering by _____, doing _____ in the system, and leaving the system by _____.
- The energy for _____ is from _____.
- The flow of energy causes _____ to occur in the system.
- The energy is entering the system by _____.
- In the system, the cycling of matter _____.
- The matter in the system enters from _____.
- When the matter leaves the system, it goes _____.

Terminology to Support Student Discourse about Phenomena

*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- | | |
|-------------------------------------|----------------------------|
| • adenosine diphosphate (ADP) | • input |
| • adenosine triphosphate (ATP) | • Krebs cycle |
| • aerobic cellular respiration | • lactic acid |
| • alcohol (ethanol) | • lactic acid fermentation |
| • alcohol fermentation | • matter cycle |
| • anaerobic cellular respiration | • mitochondria |
| • bacteria | • muscle cell |
| • carbon dioxide (CO ₂) | • NAD ⁺ |
| • cellular respiration | • NADH |
| • chemical reaction | • organic molecules |
| • cytosol | • output |
| • electron transport chain | • phosphorous (P) |
| • fermentation | • products |
| • glucose | • pyruvate (pyruvic acid) |
| • glycolysis | • reactants |
| • heterotroph | • water |
| | • yeast |

LS2 – Ecosystems: Interactions, Energy, and Dynamics

B-LS2-1. Use mathematical and/or computational representations to support explanations of biotic and abiotic factors that affect carrying capacity of ecosystems *at different scales.*

Clarification Statement: Emphasis is on quantitative analysis and comparison of the relationships among interdependent factors including boundaries, resources, climate, and challenges. Examples of mathematical comparisons could include graphs, charts, histograms, or population changes gathered from simulations and historical data sets. Examples of scales could be a pond versus an ocean.

State Assessment Boundary: Assessment does not include deriving mathematical equations to make comparisons.

Science and Engineering Practices	Disciplinary Core Ideas	Crosscutting Concepts
Using Mathematics and Computational Thinking Mathematical and computational thinking in 9-12 builds on K-8 experiences and progresses to using algebraic thinking and analysis, a range of linear and nonlinear functions including trigonometric functions, exponentials and logarithms, and computational tools for statistical analysis to analyze, represent, and model data. Simple computational simulations are created and used based on mathematical models of basic assumptions. Use mathematical and/or computational representations of phenomena or design solutions to support explanations.	LS2.A: Interdependent Relationships in Ecosystems Ecosystems have carrying capacities, which are limits to the numbers of organisms and populations they can support. These limits result from such factors as the availability of living and nonliving resources and from such challenges such as predation, competition, and disease. Organisms would have the capacity to produce populations of great size were it not for the fact that environments and resources are finite. This fundamental tension affects the abundance (number of individuals) of species in any given ecosystem.	Scale, Proportion, and Quantity The significance of a phenomenon is dependent on the scale, proportion, and quantity at which it occurs.

*Because populations can be large, students may need to understand and apply the conventions of scientific notation when working with quantities of measurement and their calculations.

Observable features of student performance by the end of the course:

1. Representation

- a. Students develop/use mathematical and/or computational models (for example: trends, averages, histograms, graphs, spreadsheets, etc.) to identify and describe the components that are relevant to support or refute an explanation of factors that affect carrying capacities of ecosystems at different scales, including:
 - i. population changes gathered from historical data and/or simulations, and
 - ii. data on:
 - 1. numbers and types of organisms,
 - 2. boundaries,
 - 3. resources, and
 - 4. climate.
- b. Students support or refute explanations about the factors (for example: boundaries resources, climate, competition, etc.) that affect the carrying capacity of an ecosystem and:
 - i. some factors have larger effects than others,
 - ii. factors are interrelated, and/or
 - iii. factor significance depends on the scale at which it occurs.

2. Mathematical and/or computational modeling

- a. Students develop/use mathematical and/or computational models (for example: trends, averages, histograms, graphs, spreadsheets, etc.) of ecosystem factors to identify changes over time in the numbers and types of organisms in ecosystems of different scales.

3. Analysis

- a. Students develop/use and analyze a mathematical and/or computational model:
 - i. To identify the interdependence of factors (both abiotic and biotic) and the resulting effect on carrying capacity.
 - ii. To predict the resulting effect on carrying capacity of a change in abiotic and/or biotic factors.
 - iii. As evidence to support or refute an explanation and identify the factors that have the largest effect on the carrying capacity of an ecosystem for an identified population.

B-LS2-1 Academic Language

Question/Sentence Stems

- The quantity of _____ and _____ can be compared.
- The proportion of _____ is _____ because _____.
- The scale of the model of _____ is _____ compared to the actual population.
- To understand the phenomenon of _____, I/we can use a scale of _____ in my/our model.

Terminology to Support Student Discourse about Phenomena

*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- | | |
|-----------------------|---------------------------|
| • abiotic factor(s) | • exponential growth |
| • abundance | • extinction |
| • autotroph | • fluctuation |
| • biodiversity | • generation |
| • biome | • heterotroph |
| • biotic factor(s) | • limiting factor |
| • carrying capacity | • limiting resource |
| • climate change | • logistic growth |
| • commensalism | • mutualism |
| • community | • niche |
| • competition | • parasitism |
| • consumer | • pollution |
| • deforestation | • population |
| • density | • population growth model |
| • density-dependent | • predation |
| • density-independent | • producer |
| • disease | • species |
| • disturbance | • stability |
| • ecosystem | • sustainable |
| • environment | • symbiosis |
| • equilibrium | • trend |

B-LS2-5. Develop a model to illustrate the role of photosynthesis and cellular respiration in the cycling of carbon among the biosphere, atmosphere, hydrosphere, and geosphere.

[State] Assessment Boundary: Assessment does not include the specific chemical steps of photosynthesis and respiration.

Science and Engineering Practices	Disciplinary Core Ideas	Crosscutting Concepts
Developing and Using Models Modeling in 9-12 builds on K-8 experiences and progresses to using, synthesizing, and developing models to predict and show how relationships among variables between systems and their components in the natural and designed worlds. Develop a model based on evidence to illustrate the relationships between systems or components of a system.	LS2.B: Cycles of Matter and Energy Transfer in Ecosystems Photosynthesis and cellular respiration are important components of the carbon cycle, in which carbon is exchanged among the biosphere, atmosphere, oceans, and geosphere through chemical, physical, geological, and biological processes. PS3.D: Energy in Chemical Processes and Everyday Life The main way that solar energy is captured and stored on Earth is through the complex chemical process known as photosynthesis. (<i>secondary</i>)	Systems and System Models Models (e.g., physical, mathematical, computer models) can be used to simulate systems and interactions—including energy, matter, and information flows—within and between systems at different scales.

Observable features of student performance by the end of the course:

1. Components of the model

- a. Students develop/use a model (conceptual, graphical, physical, etc.) and identify the relevant components, including:
 - i. inputs and outputs of photosynthesis,
 - ii. inputs and outputs of cellular respiration, and
 - iii. the biosphere, atmosphere, hydrosphere, and geosphere.

2. Relationships

- a. Students develop/use a model to identify and describe the relationships between components, including:
 - i. the exchange of carbon between organisms and the environment and
 - ii. the role of storing carbon in organisms as part of the carbon cycle.

3. Connections

- a. Students develop/use a model to describe the contributions of photosynthesis and cellular respiration to the exchange of carbon within and among the biosphere, atmosphere, hydrosphere, and geosphere.
 - i. All organisms take in matter and rearrange the atoms in chemical reactions.
 - ii. Photosynthesis captures light energy to create chemical products (limited to glucose) that can be used in cellular respiration.
 - iii. Cellular respiration is the process by which glucose reacts chemically with other compounds (limited to glycolysis, Krebs's cycle, electron transport chain, fermentation), rearranging the matter to release stored chemical energy that is used for essential life processes.
- b. Students identify the limitations of a model.

B-LS2-5 Academic Language

Question/Sentence Stems

- The key components of the system are_____.
- In the system, _____ and _____ are shown in the model.
- In the system, _____ and _____ work together to_____.
- In the system, _____ and _____ interact in _____ way.

Terminology to Support Student Discourse about Phenomena

*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- | | | |
|-------------------------------------|-------------------|------------------|
| • acidification | • combustion | • microbes |
| • atmosphere | • consumer | • mitochondrion |
| • autotroph | • decomposition | • molecule |
| • biomass | • deforestation | • ocean uptake |
| • biosphere | • Earth materials | • organic matter |
| • carbon cycle | • energy | • photosynthesis |
| • carbon dioxide (CO ₂) | • fossil fuel | • phytoplankton |
| • carbon fixation | • geosphere | • producer |
| • carbon reservoir | • glucose | • reaction |
| • carbon sink | • greenhouse gas | • respiration |
| • chloroplast | • heterotroph | • sedimentation |
| • climate change | • hydrocarbons | • sugar |
| | • hydrosphere | • transpiration |
| | • methane | • zooplankton |

B-LS2-7. Design, evaluate, and refine a solution for reducing *the impacts of human activities* on biodiversity and ecosystem health.

Science and Engineering Practices	Disciplinary Core Ideas	Crosscutting Concepts
Constructing Explanations and Designing Solutions Constructing explanations and designing solutions in 9-12 builds on K-8 experiences and progresses to explanations and designs that are supported by multiple and independent student-generated sources of evidence consistent with scientific ideas, principles, and theories. Design, evaluate, and refine a solution to a complex real-world problem, based on scientific knowledge, student-generated sources of evidence, prioritized criteria, and trade off considerations.	LS2.C: Ecosystem Dynamics, Functioning, and Resilience Moreover, anthropogenic changes (induced by human activity) in the environment—including habitat destruction, pollution, introduction of invasive species, overexploitation, and climate change—can disrupt an ecosystem and threaten the survival of some species. LS4.D: Biodiversity and Humans Biodiversity is increased by the formation of new species (speciation) and decreased by the loss of species (extinction). (<i>secondary</i>) Humans depend on the living world for the resources and other benefits provided by biodiversity. Human activity is having adverse impacts on biodiversity through overpopulation, overexploitation, habitat destruction, pollution, introduction of invasive species, and climate change. These problems have the potential to cause biological extinctions which result in decreased biodiversity and the effects may be harmful to humans and other living things. Sustaining biodiversity so that ecosystem functioning, and productivity are maintained is essential to supporting and enhancing life on Earth. Sustaining biodiversity also aids humanity by preserving landscapes of recreational or inspirational value. (<i>secondary</i>) ETS1.B: Developing Possible Solutions When evaluating solutions, it is important to take into account a range of constraints including cost, safety, reliability and aesthetics and to consider social, cultural and environmental impacts. (<i>secondary</i>)	Stability and Change Much of science deals with constructing explanations of how things change and how they remain stable.

Observable features of student performance by the end of the course:

1. Using scientific knowledge to generate design solutions

- a. Students design a solution to reduce the negative effects of human activities on the environment and biodiversity, relying on factors affecting changes and stability in biodiversity. Examples of factors include but are not limited to:
 - i. overpopulation,
 - ii. overexploitation,
 - iii. habitat destruction,
 - iv. pollution,
 - v. invasive species, and/or
 - vi. climate change.
- b. Students describe/explain how the proposed solution decreases the negative effects of human activities on the environment and biodiversity.

2. Describing criteria and constraints, including quantifications when appropriate

- a. Students describe and quantify criteria, including:
 - i. amount of reduction of impacts and/or
 - ii. human activities to be mitigated
- b. Students describe the design constraints and tradeoffs, which may include:
 - i. availability and cost of materials,
 - ii. environmental impact,
 - iii. human needs,
 - iv. safety, and/or
 - v. time (for example: construction, function, etc.).

3. Evaluating potential solutions

- a. Students evaluate the proposed solution for
 - i. cost,
 - ii. its impact on overall environmental stability and changes,
 - iii. reliability,
 - iv. safety, and/or
 - v. social, cultural, and environmental impacts.

4. Refining/optimizing the design solution

- a. Students refine the proposed solution by prioritizing the criteria and making tradeoffs to further reduce environmental impact and loss of biodiversity while addressing human needs.

B-LS2-7 Academic Language

Question/Sentence Stems

- The things that stay the same are _____.
- The things that change are _____.
- The things that are changing slowly in this system are _____.
- The _____(event) changed this system by _____.
- _____ was affected by the change of _____.
- _____ are causing this system to be unstable.

Terminology to Support Student Discourse about Phenomena

*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- | | |
|--------------------------|---------------------------|
| • abiotic | • extinction |
| • anthropogenic | • genetic diversity |
| • biodiversity | • habitat |
| • biotic | • habitat destruction |
| • captive breeding | • habitat fragmentation |
| • carrying capacity | • habitat restoration |
| • climate | • human impact |
| • climate change | • human population growth |
| • conservation | • invasive species |
| • constraints | • iterative |
| • criteria | • optimize |
| • deforestation | • overharvesting |
| • ecological restoration | • pollution |
| • ecosystem | • relative abundance |
| • ecosystem diversity | • solution |
| • ecotourism | • species diversity |
| • endangered species | • species richness |
| • environment | • sustainable development |

LS3 – Heredity: Inheritance and Variation of Traits

B-LS3-2. Make and defend a claim based on evidence that inheritable genetic variations *may result from* (1) new genetic combinations through meiosis, (2) viable errors occurring during replication, and/or (3) mutations caused by environmental factors.

Clarification Statement: Emphasis is on using data to support arguments for the way genetic variation occurs.

State Assessment Boundary: Assessment does not include the phases of meiosis or the biochemical mechanism of specific steps in the process.

Science and Engineering Practices	Disciplinary Core Ideas	Crosscutting Concepts
Engaging in Argument from Evidence Engaging in argument from evidence in 9-12 builds on K-8 experiences and progresses to using appropriate and sufficient evidence and scientific reasoning to defend and critique claims and explanations about the natural and designed world(s). Arguments may also come from current scientific or historical episodes in science. Make and defend a claim based on evidence about the natural world that reflects scientific knowledge, and student-generated evidence.	LS3.B: Variation of Traits In sexual reproduction, chromosomes can sometimes swap sections during the process of meiosis (cell division), thereby creating new genetic combinations and thus more genetic variation. Although DNA replication is tightly regulated and remarkably accurate, errors do occur and result in mutations, which are also a source of genetic variation. Environmental factors can also cause mutations in genes, and viable mutations are inherited. Environmental factors also affect expression of traits, and hence affect the probability of occurrences of traits in a population. Thus, the variation and distribution of traits observed depends on both genetic and environmental factors.	Cause and Effect Empirical evidence is required to differentiate between cause and correlation and make claims about specific causes and effects.

*References to “viable errors occurring during replication” are operationally defined as errors that bypass DNA proofreading (the cell cycle successfully moves past G₂).

Observable features of student performance by the end of the course:

1. Developing a claim

- a. Students make a claim that inheritable genetic variations may result from:
 - i. new genetic combinations through meiosis,
 - ii. viable errors occurring during replication, and/or
 - iii. mutation caused by environmental factors (mutagen).

2. Identifying scientific evidence

- a. Students identify and describe the evidence that supports the claim, including:
 - i. Variation in genetic material naturally results during meiosis from
 - 1. crossing over and
 - 2. independent assortment.
 - ii. Genetic mutations can occur due to
 - 1. replication errors and
 - 2. environmental factors (mutagen).
 - iii. Genetic material is inheritable.
- b. Students use multiple valid and reliable sources of evidence, including student generated data.

3. Evaluating and critiquing evidence

- a. Students identify strengths and weaknesses of the evidence used to support the claim, including:
 - i. types of sources,
 - ii. sufficiency, including validity and reliability, of the evidence to make and defend the claim, and/or
 - iii. any alternative interpretations of the evidence and why the evidence supports or does not support the student's claim, as opposed to any other claims.

4. Reasoning and synthesis

- a. Students use the following chain of reasoning to connect the evidence:
 - i. Genetic mutations produce genetic variation between cells or organisms.
 - ii. Genetic variations produced by mutation and meiosis can be inherited.
- b. Students use reasoning and evidence to explain that new combinations of DNA can arise from several sources, including:
 - i. meiosis,
 - ii. replication error, and/or
 - iii. mutations caused by mutagens.
- c. Students defend a claim against counterclaims by evaluating counterclaims and by describing the connections between the relevant and appropriate evidence to the strongest claim.

B-LS3-2 Academic Language

Question/Sentence Stems

- By looking at patterns in the data, I/we determined that _____ caused _____.
- _____ caused the patterns I am observing. I know this because _____.
- If _____ happens, I/we predict that _____ will occur.
- Even though I/we cannot see _____, it explains why _____ is happening.
- When I/we change _____ in the system, _____ is affected.
- The probability that _____ caused _____ is _____. I/We know this because _____.
- The evidence _____ presented in the scenario supports the claim that _____ causes _____.
- To conclude that _____ caused _____, the following evidence is needed _____.

Terminology to Support Student Discourse about Phenomena

*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- | | |
|-------------------------------|--------------------------|
| • allele | • homologous chromosome |
| • cellular division | • independent assortment |
| • centromere | • inherited |
| • chromatid | • insertion |
| • chromosome | • meiosis |
| • codon (chart) | • meiosis I |
| • crossing over | • meiosis II |
| • daughter cell | • monosomy |
| • deletion | • multicellular |
| • diploid | • mutagen |
| • deoxyribonucleic acid (DNA) | • mutation |
| • egg cell | • nondisjunction |
| • epigenetic | • offspring |
| • fertilization | • parent |
| • frameshift | • parent cell |
| • gamete | • point mutation |
| • gene | • replication |
| • gene expression | • sexual reproduction |
| • gene mutation | • somatic cell |
| • genetic code | • sperm cell |
| • genetic variation | • substitution |
| • genome | • trait |
| • haploid | • trisomy |

B-LS3-3. Apply concepts of statistics and probability to explain the variation and distribution of expressed traits in a population.

Clarification Statement: Emphasis is on the use of mathematics to describe the probability of traits as it relates to genetic and environmental factors in the expression of traits.

State Assessment Boundary: Assessment does not include Hardy-Weinberg or Chi-square analysis.

Science and Engineering Practices	Disciplinary Core Ideas	Crosscutting Concepts
Analyzing and Interpreting Data Analyzing data in 9-12 builds on K-8 experiences and progresses to introducing more detailed statistical analysis, the comparison of data sets for consistency, and the use of models to generate and analyze data. Apply concepts of statistics and probability (including determining function fits to data, slope, intercept, and correlation coefficient for linear fits) to scientific and engineering questions and problems, using digital tools when feasible.	LS3.B: Variation of Traits Environmental factors also affect expression of traits, and hence affect the probability of occurrences of traits in a population. Thus, the variation and distribution of traits observed depends on both genetic and environmental factors.	Scale, Proportion, and Quantity Algebraic thinking is used to examine scientific data and predict the effect of a change in one variable on another (e.g., linear growth vs. exponential growth).

Observable features of student performance by the end of the course:

1. Organizing data

- a. Students organize data for expressed traits in the population by:
 - i. distribution,
 - ii. frequency, and/or
 - iii. variation.

2. Identifying relationships

- a. Students analyze data (for example: probability measures) to determine the relationship between a trait's occurrence within a population and environmental factors.

3. Interpreting data

- a. Students analyze and interpret data to explain the distribution of expressed traits, including:
 - i. Recognition and use of patterns in statistical analysis to predict changes in trait distribution within a population if environmental variables change.
 - ii. Description of the expression of a chosen trait and its relationship to some environmental factors based on reliable evidence.

B-LS3-3 Academic Language

Question/Sentence Stems

- The quantity of _____ and _____ can be compared.
- The proportion of _____ is _____ because _____.
- The scale of the model of _____ is _____ compared to the actual population.
- To understand the phenomenon of _____, I/we can use a scale of _____ in my/our model.

Terminology to Support Student Discourse about Phenomena

*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- | | |
|----------------------------------|-----------------------------------|
| • allele | • incomplete dominance |
| • autosomal | • karyotype |
| • codominance | • monohybrid cross |
| • complete dominance | • parental (P) |
| • dihybrid cross | • pedigree |
| • distribution | • phenotype |
| • dominant | • phenotypic ratio |
| • environmental | • polygenic inheritance |
| • first filial (F ₁) | • population |
| • gamete | • probability |
| • gene expression | • Punnett square |
| • genetic variability | • ratio |
| • genotype | • recessive |
| • genotypic ratio | • second filial (F ₂) |
| • heredity | • sex linked |
| • heterozygous | • trait |
| • homozygous | • variation |

LS4 – Biological Evolution: Unity and Diversity

B-LS4-1. Communicate scientific information that common ancestry and biological evolution *are supported by multiple lines of empirical evidence.*

Clarification Statement: Emphasis is on students’ conceptual understanding of the role each line of evidence has relating to common ancestry and biological evolution. Examples of evidence could include similarities in DNA sequences, anatomical structures, and order of appearance of structures in embryological development.

State Assessment Boundary: Assessment is limited to conceptual explanations of the evidence for biological evolution and is not extended to the lines of evidence for specific species. Assessment does not include classification of organisms.

Science and Engineering Practices	Disciplinary Core Ideas	Crosscutting Concepts
Obtaining, Evaluating, and Communicating Information Obtaining, evaluating, and communicating information in 9-12 builds on K-8 experiences and progresses to evaluating the validity and reliability of the claims, methods, and designs. Communicate scientific information (e.g., about phenomena and/or the process of development and the design and performance of a proposed process or system) in multiple formats (including orally, graphically, textually, and mathematically).	LS4.A: Evidence of Common Ancestry and Diversity Genetic information, like the fossil record, provides evidence of evolution. DNA sequences vary among species, but there are many overlaps, notably, the ongoing branching that produces multiple lines of descent can be inferred by comparing the DNA sequences of different organisms. Such information is also derivable from the similarities and differences in amino acid sequences and from anatomical and embryological evidence. ETS2.B: Influence of Engineering, Technology, and Science on Society and the Natural World The understanding of evolutionary relationships has recently been greatly accelerated by using new molecular tools to study biology.	Patterns Different patterns may be observed at each of the scales at which a system is studied and can provide evidence for causality in explanations of phenomena.

*Because the different lines of evidence have varying scales, students may need to understand and apply the conventions of scientific notation when working with quantities of measurement and their calculations.

Observable features of student performance by the end of the course:

1. Communication

- a. Students use and cite at least two different formats (for example: oral, graphical, textual, mathematical, etc.) to communicate scientific information about how common ancestry and biological evolution are supported by multiple lines of empirical evidence.

2. Connections

- a. Students identify and communicate evidence for common ancestry and biological evolution, including:
 - i. Information derived from DNA sequences, which vary among species but have many similarities between species.
 - ii. Similarities of the patterns or amino acid sequences, even when DNA sequences are slightly different, including that multiple patterns of DNA sequences can code for the same amino acids.
 - iii. The triplet codon pattern (for example: codon chart) that is used to decode sequences of mRNA for protein construction is universal amongst known species.
 - iv. Patterns in the fossil record (for example: presence, location, and possible inferences in lines of evolutionary descent for multiple specimens, etc.).
 - v. Patterns of anatomical and embryological similarities.
- b. Students identify and communicate connections between the lines of evidence and the claim of common ancestry and biological evolution.
- c. Students communicate that together, the patterns observed at multiple spatial and temporal scales (for example: DNA sequences, embryological development, fossil record, etc.) provide evidence for relationships relating biological evolution and common ancestry.

B-LS4-1 Academic Language

Question/Sentence Stems

- I/We can observe (notice) the pattern of _____ presented in the data collected.
- I/We can observe (notice) the pattern of _____ in the data presented.
- The pattern seen in the collected data allows me/us to conclude (know) that _____.
- The observed pattern supports the conclusion that _____ is caused by _____, because _____.
- The pattern of _____ is changing over time.
- The following predictions can be made about _____ when using the pattern of _____ found in the data.

Terminology to Support Student Discourse about Phenomena

*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- amino acid sequencing
- analogous structure
- anatomy
- biochemical evidence
- biogeography
- cladogram
- common ancestry
- comparative anatomy
- descent with modification
- DNA sequencing
- electrophoresis
- embryo
- embryology
- evolution
- evolutionary tree
- fossil record
- heritable trait
- homologous structure
- homology
- natural selection
- paleontology
- phenotypic similarity
- phylogenetic tree
- phylogeny
- sedimentary layers
- species
- vestigial structure

B-LS4-2. Construct an explanation based on evidence that the process of evolution *primarily results from* four factors: (1) the potential for a species to increase in number, (2) the heritable genetic variation of individuals in a species due to mutation and sexual reproduction, (3) competition for limited resources, and (4) the proliferation of those organisms that are better able to survive and reproduce in the environment.

Clarification Statement: Emphasis is on using evidence to explain the influence each of the four factors has on the number of organisms, behaviors, morphology, or physiology in terms of ability to compete for limited resources and subsequent survival of individuals and adaptation of species. Examples of evidence could include mathematical models such as simple distribution graphs and proportional reasoning.

State Assessment Boundary: Assessment does not include other mechanisms of evolution, such as genetic drift, gene flow through migration, and co-evolution.

Science and Engineering Practices	Disciplinary Core Ideas	Crosscutting Concepts
Constructing Explanations and Designing Solutions Constructing explanations and designing solutions in 9-12 builds on K-8 experiences and progresses to explanations and designs that are supported by multiple and independent student-generated sources of evidence consistent with scientific ideas, principles, and theories. Construct an explanation based on valid and reliable evidence obtained from a variety of sources (including [models, peer review, simulations, theories, students' own investigations]) and the assumption that theories and laws that describe the natural world operate today as they did in the past and will continue to do so in the future.	LS4.B: Natural Selection Natural selection occurs only if there is both (1) variation in the genetic information between organisms in a population and (2) variation in the expression of that genetic information—that is, trait variation—that leads to differences in performance among individuals. LS4.C: Adaptation Evolution is driven by the interaction of four factors: (1) the potential for a species to increase in number, (2) the genetic variation of individuals in a species due to mutation and sexual reproduction, (3) competition for an environment's limited supply of the resources that individuals need in order to survive and reproduce, and (4) the ensuing proliferation of those organisms that are better able to survive and reproduce in that environment.	Cause and Effect Empirical evidence is required to differentiate between cause and correlation and make claims about specific causes and effects.

Observable features of student performance by the end of the course:

1. Articulating the explanation of phenomena

- a. Students articulate a statement that evolution is caused primarily by one or more of the four factors:
 - i. potential of species to increase in number,
 - ii. heritable genetic variation of individuals in a species is due to mutation and sexual reproduction,
 - iii. competition for limited resources, and/or
 - iv. fitness.

2. Evidence

- a. Students identify and describe evidence necessary to construct an explanation, including:
 - i. As a species grows in number, competition for limited resources can arise.
 - ii. Individuals in a species have genetic variation that is passed on to their offspring.
 - iii. Individuals can have specific traits that increase fitness relative to other individuals in the species.
- b. Students use multiple reliable and valid sources, including student generated data.

3. Reasoning

- a. Students use the following chain of reasoning to connect the evidence for the explanation:
 - i. Genetic variation can lead to variation of expressed traits in individuals in a population.
 - ii. Individuals with traits that give competitive advantages have increased fitness.
 - iii. Individuals with increased fitness will provide their specific genetic variations to a greater proportion of the next generation.
 - iv. Over many generations, groups of individuals with particular traits that increase fitness in distinct environments using distinct resources can speciate.
- b. Students use the evidence to describe the following in their explanation:
 - i. The difference between natural selection and biological evolution.
 - ii. The cause-and-effect relationship between genetic variation, the selection of traits that increase fitness, and the evolution of populations that express the trait.

B-LS4-2 Academic Language

Question/Sentence Stems

- By looking at patterns in the data, I/we determined that _____ caused _____.
- _____ caused the patterns I am observing. I know this because _____.
- If _____ happens, I/we predict that _____ will occur.
- Even though I/we cannot see _____, it explains why _____ is happening.
- The evidence _____ presented in the scenario supports the claim that _____ causes _____.
- To conclude that _____ caused _____, the following evidence is needed _____.

Terminology to Support Student Discourse about Phenomena

*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- adaptation
- behavior
- beneficial
- competition
- detrimental
- evolution
- fitness
- gene pool
- genetic variation
- geographic isolation
- heritable
- limited resources
- morphology
- mutation
- natural selection
- phenotypic expression
- physiology
- population
- proliferation
- reproductive isolation
- selective pressure
- sexual reproduction
- sexual selection
- speciation
- species
- trait

B-LS4-4. Construct an explanation based on evidence for how natural selection *leads to* adaptation of populations.

Clarification Statement: Emphasis is on using data to provide evidence for how specific biotic and abiotic differences in ecosystems (such as ranges of seasonal temperature, long-term climate change, acidity, light, geographic barriers, or evolution of other organisms) contribute to a change in gene frequency over time, leading to adaptation of populations.

State Assessment Boundary: Assessment does not include allele frequency calculations.

Science and Engineering Practices	Disciplinary Core Ideas	Crosscutting Concepts
Constructing Explanations and Designing Solutions Constructing explanations and designing solutions in 9-12 builds on K-8 experiences and progresses to explanations and designs that are supported by multiple and independent student-generated sources of evidence consistent with scientific ideas, principles, and theories. Construct an explanation based on valid and reliable evidence obtained from a variety of sources (including [models, peer review, simulations, theories, students' own investigations]) and the assumption that theories and laws that describe the natural world operate today as they did in the past and will continue to do so in the future.	LS4.C: Adaptation Natural selection leads to adaptation, that is, to a population dominated by organisms that are anatomically, behaviorally, and physiologically well suited to survive and reproduce in a specific environment. That is, the differential survival and reproduction of organisms in a population that have an advantageous heritable trait leads to an increase in the proportion of individuals in future generations that have the trait and to a decrease in the proportion of individuals that do not.	Cause and Effect Empirical evidence is required to differentiate between cause and correlation and make claims about specific causes and effects.

Observable features of student performance by the end of the course:

1. Articulating the explanation of phenomena

- a. Students articulate a statement that describes/explains the cause-and-effect relationship between natural selection and adaptation.

2. Evidence

- a. Students identify and describe the evidence necessary to construct an explanation, including:
 - i. Changes in a population when some feature of the environment changes.
 - ii. Relative survival rates of organisms with different traits in a specific environment.
 - iii. Individuals in a species have genetic variation that is passed on to their offspring.
 - iv. Specific traits can increase an individual's fitness relative to other individuals in the species.
- b. Students use multiple reliable and valid sources, including student generated data.

3. Reasoning

- a. Students use the following chain of reasoning to synthesize the valid and reliable evidence about how natural selection provides a mechanism for species to adapt to changes in their environment:
 - i. Abiotic and biotic differences in ecosystems contribute to changes in gene and/or allele frequency over time through natural selection.
 - ii. Increasing gene and/or allele frequency in a population results in an increasing fraction of the population in each successive generation that carries a particular gene and expresses a particular trait.
 - iii. Over time, this process leads to a population that is adapted to a particular environment by the widespread expression of a trait that confers a competitive advantage in that environment.
 - iv. If abiotic or biotic factors of that environment change, then there is the potential for a change in gene and/or allele frequency again.

B-LS4-4 Academic Language

Question/Sentence Stems

- By looking at patterns in the data, I/we determined that _____ caused _____.
- _____ caused the patterns I am observing. I know this because _____.
- If _____ happens, I/we predict that _____ will occur.
- Even though I/we cannot see _____, it explains why _____ is happening.
- The probability that _____ caused _____ is _____. I/We know this because _____.
- The evidence _____ presented in the scenario supports the claim that _____ causes _____.
- To conclude that _____ caused _____, the following evidence is needed _____.

Terminology to Support Student Discourse about Phenomena

*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- abiotic
- adaptation
- advantageous trait
- allele frequency
- beneficial
- biotic
- coevolution
- convergent evolution
- detrimental
- distribution
- diverge
- ecosystem
- fitness
- gene
- gene frequency
- gene pool
- genetic variation
- geographic isolation
- mutation
- natural selection
- phenotypic variation
- population
- survival rate
- trait
- variation

B-LS4-5. Evaluate the evidence supporting claims that changes in environmental conditions *may result in* (1) increases in the number of individuals of some species, (2) the emergence of new species over time, and (3) the extinction of other species.

Clarification Statement: Emphasis is on determining cause and effect relationships for how changes to the environment such as deforestation, fishing, application of fertilizers, drought, flood, and the rate of change of the environment affect distribution or disappearance of traits in species.

Science and Engineering Practices	Disciplinary Core Ideas	<i>Crosscutting Concepts</i>
Engaging in Argument from Evidence Engaging in argument from evidence in 9-12 builds on K-8 experiences and progresses to using appropriate and sufficient evidence and scientific reasoning to defend and critique claims and explanations about the natural and designed world(s). Arguments may also come from current or historical episodes in science. Evaluate the evidence behind currently accepted explanations or solutions to determine the merits of arguments.	LS4.C: Adaptation Changes in the physical environment, whether naturally occurring or human induced, have thus contributed to the expansion of some species, the emergence of new distinct species as populations diverge under different conditions, and the decline—and sometimes the extinction—of some species. Species become extinct because they can no longer survive and reproduce in their altered environment. If members cannot adjust to change that is too fast or drastic, the opportunity for the species' evolution is lost.	Cause and Effect Empirical evidence is required to differentiate between cause and correlation and make claims about specific causes and effects.

Observable features of student performance by the end of the course:

1. Identifying the explanation and supporting evidence

- a. Students identify an explanation about the impacts of changes in environmental conditions on species and the relevant evidence including:
 - i. increases in the number of individuals of some species,
 - ii. emergence of a new species over time, and/or
 - iii. extinction of a species.

2. Identifying additional evidence relevant to the evaluation

- a. Students identify and describe additional evidence (for example: data, information, etc.) that was not provided but is relevant to the explanation and its evaluation including:
 - i. Data indicating the change over time in:
 1. the number of individuals in each species,
 2. the number of species in an environment, and/or
 3. the environmental conditions.
 - ii. Environmental factors that can determine an individual's fitness.

3. Evaluating and critiquing

- a. Students use additional evidence to assess the validity, reliability, strengths, and weaknesses of the evidence and evaluate its ability to support logical and reasonable arguments about:
 - i. environmental changes,
 - ii. changes in numbers of individuals in each species,
 - iii. the number of species in an environment, and/or
 - iv. the emergence or extinction of species.
- b. Students evaluate the evidence for the degree to which it supports relationships between:
 - i. environmental changes,
 - ii. changes in numbers of individuals in each species,
 - iii. the number of species in an environment, and/or
 - iv. the emergence or extinction of species.
- c. Students evaluate the degree to which the empirical evidence can be used to construct logical arguments that identify relationships between environmental changes and changes in the numbers of individuals or species based on environmental factors that can determine the fitness of individuals in a species.

B-LS4-5 Academic Language

Question/Sentence Stems

- By looking at patterns in the data, I/we determined that _____ caused _____.
- _____ caused the patterns I am observing. I know this because _____.
- If _____ happens, I/we predict that _____ will occur.
- Even though I/we cannot see _____, it explains why _____ is happening.
- The probability that _____ caused _____ is _____. I/We know this because _____.
- The evidence _____ presented in the scenario supports the claim that _____ causes _____.
- To conclude that _____ caused _____, the following evidence is needed _____.

Terminology to Support Student Discourse about Phenomena

*Teaching words or concepts in isolation or prior to experiences that give context (frontloading) deprives students of sense-making opportunities that lead to a greater depth of conceptual understanding.

- abiotic
- abiotic factor
- adaptation
- advantageous
- allele frequency
- biotic
- biotic factor
- detrimental
- distribution
- disturbance
- diverge
- ecosystem
- environment
- evolution
- extinction
- fitness
- fossil
- founder effect
- gene
- gene flow
- gene frequency
- gene pool
- genetic drift
- genetic variation
- geographic isolation
- geologic record
- gradualism
- natural selection
- phenotypic variation
- population
- selective pressure
- speciation
- species
- stability
- survival rate
- trait
- variation

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Appendix A: Biology 1 Vertical Articulation of Disciplinary Core Idea Instructional Focus Statements

This appendix presents the vertical articulation of Disciplinary Core Idea (DCI) Aligned Instructional Focus statements found in the Vertical Articulations on the [Science Instructional Resources webpage](#). It is intended to help educators understand how conceptual understanding develops across grade levels and courses and how current learning connects to both prior and future instruction.

This appendix is designed to support instructional planning, curriculum alignment, and understanding of the broader progression of science concepts across grades and courses.

Instructional Focus statements clarify content expectations associated with a Performance Expectation but should be considered alongside the associated Science and Engineering Practices (SEPs) and Crosscutting Concepts (CCCs).

The appendix is organized by Performance Expectation (PE) and associated DCI, using the following format:

Domain Core Idea

Performance Expectation

Disciplinary Core Idea

Previous Learning	Current Grade	Future Learning
Related DCI Aligned Instructional Focus statements from earlier grades are presented in this column to illustrate the conceptual foundations of the current learning.	DCI Aligned Instructional Focus statements for the current grade are presented in this column. If a statement aligns to more than one PE, the applicable PE code(s) follows the statement. Only relevant Extended Knowledge statements are included.	- Related DCI Aligned Instructional Focus statements from future grades and courses are presented in this column to illustrate how the concept develops over time. - If the articulation for the DCI ends in Biology 1, this box will be absent.

The appendix is not intended to represent a complete curriculum or pacing guide. Rather, it provides a conceptual map of how the knowledge and understandings associated with a DCI progress across grades, courses, and content.

LS1 – From Molecules to Organisms: Structures and Processes

B-LS1-1. Construct an explanation based on evidence for how the structure of DNA determines the structure of proteins, which carry out the essential functions of life through systems of specialized cells.

LS1.A: Structure and Function

Sixth Grade	Biology 1	Biology 2
Interpret the roles and interactions of major cell structures (membrane, wall, nucleus, chloroplasts, mitochondria) in supporting whole-cell function, without describing molecular or chemical mechanisms.	- Describe the structure and functions of DNA. - Describe proteins as the primary functional products of genetic information that enable cells and tissues to perform specialized functions. - Apply the pathway from DNA in the nucleus to protein production by summarizing transcription to mRNA, polypeptide assembly at ribosomes, and protein modification and packaging by the ER and Golgi apparatus. - Explain that all cells in an organism contain the same DNA but can produce different sets of proteins that give rise to specialized cell functions. - Analyze how changes in a gene's information can change a protein and alter cell or tissue function at a conceptual level. <i>Extended Knowledge</i> - Recognize that not all regions of DNA directly code for proteins. - Understand that gene structure and RNA processing affect how genetic information is used.	- Differentiate coding and non-coding DNA and relate non-coding regions to organizational or regulatory roles. - Describe introns and exons in eukaryotic genes and explain how RNA processing produces mature mRNA, allowing a single gene to support multiple functional outcomes. - Outline major stages of transcription and translation and the roles of key cellular structures involved, without reaction mechanisms or kinetics.

LS3.A: Inheritance of Traits

Eighth Grade	Biology 1	Biology 2
- Describe that genes are located on chromosomes, which occur in pairs carrying two alleles. - Explain that a gene can direct the production of a specific protein and that protein function influences traits, recognizing that many traits involve multiple components. - Describe mutations as changes in genes that can lead to changes in proteins, with effects that may be beneficial, harmful, or neutral. - Explain that inherited trait variation arises from the subset of chromosomes (and alleles) an offspring inherits.	- Explain that gene's nucleotide sequence is used to make proteins through transcription of DNA into mRNA and translation of mRNA into a polypeptide. - Explain how traits are inherited and expressed, using central dogma reasoning. - Explain that changes in a gene's sequence can result in changes to a protein and lead to differences in traits. <i>Extended Knowledge</i> - Recognize that not all DNA codes directly for proteins and that some regions influence how genes are used. - Understand that some traits are influenced by multiple genes (for example, eye color, height, disease susceptibility, mammal skin tone, etc.) and may show a range of forms.	- Explain that a gene's nucleotide sequence specifies a protein or a functional RNA product. - Explain how changes in DNA sequence can alter proteins or RNAs and thereby influence traits at the organism level. - Identify that some DNA does not code for proteins and plays regulatory or structural roles that influence gene expression. - Describe how regulatory DNA regions can affect when, where, or how much a gene is expressed. - Recognize that epigenetic influences can alter gene expression without changing the nucleotide sequence. - Explain that many traits are polygenic, leading to complex genotype–phenotype relationships.

B-LS1-4. Use a model to illustrate the role of cellular division (mitosis) [sic] and differentiation in producing and maintaining complex organisms.

LS1.B: Growth and Development of Organisms

Eighth Grade	Biology 1
- Analyze how characteristic reproductive behaviors in animals increase the probability of successful reproduction. - Explain how specialized reproductive structures in plants, including interactions with animals, support pollination and seed dispersal. - Explain how genetic factors (without detailing inheritance mechanisms) and local environmental conditions together influence patterns of growth and typical adult size ranges within a species.	- Summarize the outcomes of major stages of the cell cycle and mitosis in terms of how genetic continuity is maintained. - Explain that daughter cells produced by mitosis receive identical genetic information, supporting growth and tissue replacement. - Differentiate general features of mitosis in plant and animal cells. - Explain how mitosis and cell differentiation together support organismal growth, repair, and maintenance by preserving genetic information. - Explain at a conceptual level that cell-cycle regulation helps maintain normal growth and that loss of regulation can result in uncontrolled cell division (for example, cancer). - Evaluate how changes to genetic information or regulation can influence differentiation outcomes and tissue function. <i>Extended Knowledge</i> - Recognize that additional regulatory processes influence cell division beyond those required to explain normal growth and repair.

B-LS1-5 Use a model to illustrate how photosynthesis transforms light energy into stored chemical energy.

LS1.C: Organization for Matter and Energy Flow in Organisms

Seventh Grade	Biology 1
• Recognize that plants, algae, and many microorganisms carry out photosynthesis. • Explain that photosynthetic organisms use light energy, carbon dioxide, and water to make sugars and release oxygen. • Describe how sugars made by producers supply matter and energy for producers and consumers.	• Explain that chemical reactions within living systems rearrange matter and transfer energy, conserving atoms as energy flows through the system. (B-LS1-6, B-LS1-7) • Explain how photosynthesis stores light energy as chemical energy in sugars. • Explain that sugars supply both matter and energy for building molecules and supporting life functions. (B-LS1-6, B-LS1-7) <i>Extended Knowledge</i> • Recognize that additional biological processes and structures support photosynthesis and cellular respiration beyond those needed to explain matter and energy flow at a system level. (B-LS1-7) • Understand that different biological strategies for acquiring and releasing energy exist, without describing biochemical pathways or reaction mechanisms. (B-LA1-7)

B-LS1-6. Construct and revise an explanation based on evidence for how carbon, hydrogen, and oxygen from sugar molecules may combine with other elements to form amino acids and other large carbon-based molecules necessary for essential life processes.

LS1.C: Organization for Matter and Energy Flow in Organisms

Seventh Grade	Biology 1
- Describe how sugars made by producers supply matter and energy for producers and consumers. - Explain that food is broken down and rearranged inside organisms to support growth or release energy, with matter conserved and energy transferred during these chemical reactions.	- Explain that chemical reactions within living systems rearrange matter and transfer energy, conserving atoms as energy flows through the system. (B-LS1-5, B-LS1-7) - Explain that sugars supply both matter and energy for building molecules and supporting life functions. (B-LS1-5, B-LS1-7) - Describe how major biomolecules, including carbohydrates, lipids, proteins, and nucleic acids, function to store energy, build cellular structures, and support essential life processes.

B-LS1-7. Use a model to illustrate that cellular respiration is a chemical process whereby the bonds of food molecules are broken and the bonds in new compounds are formed, resulting in a net transfer of energy.

LS1.C: Organization for Matter and Energy Flow in Organisms

Seventh Grade	Biology 1
- Describe how sugars made by producers supply matter and energy for producers and consumers. - Explain that food is broken down and rearranged inside organisms to support growth or release energy, with matter conserved and energy transferred during these chemical reactions.	- Explain that chemical reactions within living systems rearrange matter and transfer energy, conserving atoms as energy flows through the system. (B-LS1-5, B-LS1-6) - Explain that sugars supply both matter and energy for building molecules and supporting life functions. (B-LS1-5, B-LS1-6) - Describe how major biomolecules, including carbohydrates, lipids, proteins, and nucleic acids, function to store energy, build cellular structures, and support essential life processes. - Describe that cellular respiration rearranges sugars and oxygen to release energy for essential cell functions. - Compare aerobic and anaerobic respiration in terms of inputs, outputs, and relative energy yield. <i>Extended Knowledge</i> - Recognize that additional biological processes and structures support photosynthesis and cellular respiration beyond those needed to explain matter and energy flow at a system level. (B-LS1-5) - Understand that different biological strategies for acquiring and releasing energy exist, without describing biochemical pathways or reaction mechanisms. (B-LS1-5)

LS2 – Ecosystems: Interactions, Energy, and Dynamics

B-LS2-1. Use mathematical and/or computational representations to support explanations of biotic and abiotic factors that affect carrying capacity of ecosystems at different scales.

LS2.A: Interdependent Relationships in Ecosystems

Seventh Grade	Biology 1	Biology 2
- Define resources (biotic and abiotic) and explain that access to resources limits organism and population growth and reproduction. - Relate changes in resource availability to changes in organism growth and population size. - Explain how organisms with similar resource requirements compete for limited resources, constraining population size. - Describe common interaction types; competition, predation, and mutualism; and the general population trends each tends to produce. - Explain that similar interaction patterns occur across ecosystems even when the species involved differ.	- Define carrying capacity as the limit on population size set by finite resources and biotic challenges (predation, competition, disease). - Describe how limits from resources and biotic challenges affect species abundance within defined ecosystem boundaries. - Explain how changes in a limiting factor (for example, decreased prey or increased disease) can shift carrying capacity. - Describe overshoot and subsequent dieback qualitatively, as temporary population increases beyond resource limits followed by reduction toward sustainable levels. <i>Extended Knowledge</i> - Understand that population responses to limits vary with the nature and timing of environmental changes. - Recognize that ecosystems operate at different spatial and temporal scales, and that populations at one scale can influence resources or conditions at another.	- Describe carrying capacity as a function of finite resources and biotic challenges, and explain how this constrains population abundance. - Explain that abundance reflects the tension between potential population growth and environmental limits. - Differentiate outcomes associated with modest versus large disturbances, relating disturbance magnitude to expected population and ecosystem responses.

B-LS2-5. Develop a model to illustrate the role of photosynthesis and cellular respiration in the cycling of carbon among the biosphere, atmosphere, hydrosphere, and geosphere.

LS2.B: Cycles of Matter and Energy Transfer in Ecosystems

Fifth, Sixth, and Seventh Grade	Biology 1	Biology 2
Fifth Grade (ESS2.A Earth Materials and Systems) - Identify Earth’s major systems as the geosphere, hydrosphere, atmosphere, and biosphere. - Describe examples of interactions among Earth’s systems. - Relate interactions among Earth’s systems to changes in landforms, movement of water, and living things. Sixth Grade (ESS) - Explain how water cycles among land, oceans, and the atmosphere, including changes in form and location over time. - Identify sunlight and gravity as the primary factors that drive water movement and cycling on Earth. Seventh Grade (LS2.B: Cycles of Matter and Energy Transfer in Ecosystems) - Explain how matter and energy move through food webs in ecosystems. - Explain that matter and atoms cycle between living organisms and nonliving parts of ecosystems, including transfers to and from the physical environment. - Explain that decomposers recycle materials from dead organisms back to soil or water for reuse by other organisms.	- Describe the carbon cycle as the movement of carbon among organisms and between the biosphere, atmosphere, hydrosphere, and geosphere. - Explain how photosynthesis, cellular respiration, decomposition, and combustion connect organisms and environments through carbon exchange. <i>Extended Knowledge</i> - Recognize that carbon cycling can occur over different time scales, without analyzing rates or residence times. - Understand that carbon cycling is one example within broader biogeochemical cycles (e.g., water, nitrogen, phosphorus).	- Explain that photosynthesis provides stored chemical energy that fuels cellular respiration and supports ecosystem-level energy flow. - Describe that all organisms carry out cellular respiration (aerobic or anaerobic) to release chemical energy for life processes and return matter to the environment. - Explain that as matter and energy move through food webs, only a fraction becomes biomass at each successive trophic level, and this inefficiency limits the number of trophic levels in ecosystems. - Recognize that matter (atoms) is conserved as it moves through food webs and is recombined into different molecules, while energy flows through ecosystems and is released to the environment.

PS3.D: Energy in Chemical Processes and Everyday Life

Seventh Grade	Biology 1
• Explain that photosynthesis is a chemical process that requires energy input from sunlight to form sugars and release oxygen. • Describe that cellular respiration is a chemical process that uses oxygen to release stored chemical energy from sugars and produces carbon dioxide. • Demonstrate that energy flows qualitatively from sunlight → stored chemical energy in sugars → usable energy released during respiration.	• Communicate how photosynthesis is the primary process by which solar energy is captured and stored as chemical energy in sugars on Earth. • Trace energy flow qualitatively from the Sun to stored chemical energy in sugars and into biological systems within the carbon cycle. • Recognize that stored chemical energy in sugars is released during cellular respiration and can be used to form macromolecules that support organismal structure and function.

B-LS2-7. Design, evaluate, and refine a solution for reducing the impacts of human activities on biodiversity and ecosystem health.

LS2.C: Ecosystem Dynamics, Functioning, and Resilience

Seventh Grade	Biology 1	Biology 2
• Explain that changes to physical or biological components can shift multiple populations within an ecosystem. • Define biodiversity as the variety of species in ecosystems and recognize it as a commonly used indicator of ecosystem health. • Describe examples of ecosystem services that benefit people (for example, water purification, nutrient recycling, pollination, erosion prevention, climate regulation).	• Identify human activities that disrupt ecosystems, including habitat destruction, pollution, introduction of invasive species, overexploitation, and long-term changes in climate patterns. • Describe how these activities alter environmental conditions or resource availability within ecosystems. • Explain how such changes can limit organism survival or reproduction, threatening the persistence of some species. • Describe that actions such as habitat restoration, invasive-species management, pollution reduction, and sustainable harvest practices can improve conditions that support ecosystem functioning. <i>Extended Knowledge</i> • Recognize that species with specialized habitat requirements may be more vulnerable to environmental changes. • Understand that ecosystem recovery following disturbance can vary depending on the extent of change.	• Describe that under relatively stable conditions, complex interactions can keep ecosystem structure and population sizes within identifiable ranges. • Explain that ecosystems may return to prior states following modest disturbances when key species, resources, and interactions remain intact (resilience). • Explain that extreme disturbances or large population changes can disrupt resources and habitat availability, resulting in ecosystems with different structures and interactions. • Describe factors that influence biodiversity, including resource availability, species interactions, and physical environmental conditions. • Explain that ecosystem stability depends on both the magnitude of environmental change and the relationships among organisms and conditions.

LS4.D: Biodiversity and Humans

Seventh Grade	Biology 1	Biology 2
- Explain that humans depend on functioning ecosystems to maintain access to natural resources. - Explain that changes in biodiversity can influence the availability of resources humans rely on, including food, energy, and medicines. - Describe how biodiversity supports ecosystem services that benefit humans, such as water purification and nutrient recycling. - Explain how reductions in biodiversity can limit ecosystem services.	- Explain that humans depend on biodiversity for resources and benefits provided by living systems. - Describe how human activities such as habitat destruction, overexploitation, pollution, invasive species introduction, and long-term changes in climate patterns can reduce biodiversity. - Explain how reductions in biodiversity can disrupt ecosystem functioning and productivity, affecting humans and other organisms. - Explain why sustaining biodiversity supports continued ecosystem functioning. - Describe tradeoffs between short-term human benefits and long-term impacts on biodiversity and ecosystem functioning. <i>Extended Knowledge</i> - Recognize that biodiversity can vary across ecosystems and regions.	- Explain how evolutionary processes such as adaptation, speciation, and extinction contribute to changes in biodiversity over time. - Explain the consequences of biodiversity loss for ecosystem resilience and stability.

ETS1.B: Developing Possible Solutions

Seventh Grade	Biology 1	Biology 2
- Classify features of competing versions by how consistently they align with stronger outcomes across the listed conditions. - Explain why one feature should be kept or removed using patterns from the data. - Analyze two feasible options and name the trade-off that drives the next revision choice.	- Identify criteria for reducing a biodiversity impact and list the constraints (for example, cost, safety, reliability, and aesthetics, and consider social, cultural, and environmental impacts, etc.). - Compare feasible options by classifying how each meets the prioritized criteria and respects constraints, including social, cultural, and environmental effects. - Identify the single most limiting constraint for this problem and explain how it should inform the next revision choice. - Explain which option best fits the criteria using evidence from biodiversity outcomes. - Analyze how one targeted refinement would improve outcomes and name the trade-off accepted to keep within constraints. <i>Extended Knowledge</i> - Recognize when combining one feature from each of two options would better balance benefits and impacts and name the feature pair to keep.	- Identify criteria and constraints (for example, cost, safety, reliability, and aesthetics, and to consider social, cultural, and environmental impacts, etc.) for the biodiversity problem in clear, measurable terms. - Classify proposed solutions by the features that align with prioritized criteria and by anticipated social, cultural, and environmental effects. - Analyze how a small refinement that improves fit to the criteria while acknowledging one trade-off (for example cost, reliability, access).

LS3 – Heredity: Inheritance and Variation of Traits

B-LS3-2. Make and defend a claim based on evidence that inheritable genetic variations may result from (1) new genetic combinations through meiosis, (2) viable errors occurring during replication, and/or (3) mutations caused by environmental factors.

LS3.B: Variation of Traits

Eighth Grade	Biology 1	Biology 1
- Explain that, in sexual reproduction, offspring inherit two alleles per gene (one from each parent), which may be identical or different, contributing to variation among individuals. - Explain that mutations are changes in genetic information that can affect proteins and traits, with effects that may be beneficial, harmful, or neutral.	- Explain how independent assortment and crossing over during meiosis contribute to new combinations of alleles. - Explain that mutations can introduce new genetic variants that may be inherited. - Differentiate sources of mutation (replication errors and environmental factors) and explain how the location and type of change can influence trait effects. - Explain that environmental factors can influence how traits are expressed and affect observed patterns of traits in populations. (B-LS3-3) - Explain how combinations of alleles can account for patterns of trait variation and distribution in populations. (B-LS3-3) <i>Extended Knowledge</i> - Recognize that some traits are influenced by multiple genes and may show a range of forms in a population. (B-LS3-3) - Identify that environmental factors can influence trait expression without changing DNA sequence, including epigenetic influences. (B-LS3-3)	- Explain how genetic factors (such as multiple loci) and environmental factors (such as resource gradients) together shape the distribution of traits in populations. - Explain how recombination and mutation provide heritable variation, while environmental influences on trait expression (not inheritance) modulate observed traits across time and conditions.

B-LS3-3. Apply concepts of statistics and probability to explain the variation and distribution of expressed traits in a population.

LS3.B: Variation of Traits

Eighth Grade	Biology 1	Biology 2
Explain that, in sexual reproduction, offspring inherit two alleles per gene (one from each parent), which may be identical or different, contributing to variation among individuals.	- Explain that environmental factors can influence how traits are expressed and affect observed patterns of traits in populations. (B-LS3-2) - Explain how combinations of alleles can account for patterns of trait variation and distribution in populations. (B-LS3-2) <i>Extended Knowledge</i> - Recognize that some traits are influenced by multiple genes and may show a range of forms in a population. (B-LS3-2) - Identify that environmental factors can influence trait expression without changing DNA sequence, including epigenetic influences. (B-LS3-2)	- Explain how genetic factors (such as multiple loci) and environmental factors (such as resource gradients) together shape the distribution of traits in populations. - Explain how recombination and mutation provide heritable variation, while environmental influences on trait expression (not inheritance) modulate observed traits across time and conditions.

LS4 – Biological Evolution: Unity and Diversity

B-LS4-1. Communicate scientific information that common ancestry and biological evolution are supported by multiple lines of empirical evidence.

LS4.A: Evidence of Common Ancestry and Diversity

Eighth Grade	Biology 1	Biology 2
- Describe the fossil record as a chronological collection of fossils. - Use the fossil record to identify patterns of existence, diversity, extinction, and change in life forms over time. - Compare fossils from different time periods to identify similarities and differences among organisms. - Compare anatomical structures among modern organisms and between modern and fossil organisms. - Use anatomical and fossil evidence to describe historical relationships among organisms.	- Use multiple lines of evidence, including fossils, comparative anatomy, genetics and molecular biology, and embryology, to support claims that species share common ancestry. - Explain how similarities and differences in DNA sequences provide evidence consistent with shared ancestry among organisms. - Explain how similarities and differences in amino acid sequences provide evidence of relatedness among organisms. - Describe how anatomical and embryological features are used as evidence to support common ancestry. - Describe evolutionary relationships as branching patterns that represent lines of common ancestry <i>Extended Knowledge</i> - Recognize that different types of evidence contribute different kinds of information about evolutionary relationships.	- Explain how multiple lines of evolutionary evidence, including fossil, molecular, anatomical, and embryological data, are compared for their consistency and explanatory power. - Explain how molecular sequence data can indicate degrees of relatedness and resolve relationships among organisms. - Describe branching patterns as models of divergent evolutionary lineages, including the assumptions and limitations of these representations. - Explain how evidence from different sources can be combined to strengthen, refine, or challenge claims about common ancestry.

ETS2.B: Influence of Engineering, Technology, and Science on Society and the Natural World

Seventh Grade	Biology 1	Biology 2
- Sort examples of human activity by their short- and long-term consequences, positive or negative, for people and for the natural environment. - Explain how scientific understanding of matter, energy, or environmental processes can change which technologies are adopted or phased out. - Evaluate how two mitigation technologies differ in effectiveness based on regional resources and societal priorities.	- Identify a molecular technology used to determine relatedness among organisms and note the biological question it addresses. - Compare two molecular approaches by classifying the kind of information each provides for inferring evolutionary relationships using qualitative terms only. - Explain how using molecular evidence can clarify or reorganize groups compared with classifications based only on observable traits. - Evaluate one practical implication of adopting molecularly informed classifications for society or the natural world (for example conservation targeting, invasive-species recognition, disease tracking) and state a trade-off accepted in that choice. <i>Extended Knowledge</i> - Identify a situation where molecular grouping would likely change a real-world decision (for example which population to conserve first) and explain.	- Explain how advances in technology have expanded the types of evidence available to support evolutionary explanations. - Explain how new technologies can lead to changes in scientific understanding of evolutionary relationships. - Explain how changes in available evidence can lead to revisions of scientific models of evolutionary history. - Describe how the interpretation and use of evolutionary evidence can influence decisions in areas such as conservation, medicine, or biotechnology.

B-LS4-2. Construct an explanation based on evidence that the process of evolution primarily results from four factors: (1) the potential for a species to increase in number, (2) the heritable genetic variation of individuals in a species due to mutation and sexual reproduction, (3) competition for limited resources, and (4) the proliferation of those organisms that are better able to survive and reproduce in the environment.

LS4.B: Natural Selection

Eighth Grade	Biology 1	Biology 2
- Explain that natural selection can cause certain traits to become more common in a population over many generations. - Relate changes in trait prevalence to differences in survival and reproductive success under specific environmental conditions. - Explain how environmental factors influence which inherited traits persist in populations.	- Explain that natural selection occurs only when variation exists among individuals within a population. - Explain that inherited differences in traits can lead to differences in survival or reproductive success under specific environmental conditions. - Explain how differential survival and reproduction among individuals provide the conditions necessary for natural selection to operate in populations. <i>Extended Knowledge</i> - Recognize that not all genetic differences result in observable trait differences.	- Explain how inherited variation and environmental pressures interact to shape changes in trait distributions across generations. - Explain how population-level patterns can be interpreted to determine whether observed changes are consistent with natural selection. - Use the Hardy–Weinberg model to explain how stable allele frequencies reflect equilibrium conditions and how deviations from expected frequencies can indicate evolutionary influences such as natural selection.

LS4.C: Adaptation

Eighth Grade	Biology 1	Biology 2
• Explain that adaptation occurs over many generations when natural selection favors traits that improve survival and reproduction under specific environmental conditions (fitness). • Describe adaptation as a population-level process rather than a change within an individual organism's lifetime. • Explain that traits supporting successful survival and reproduction tend to become more common in a population over time. • Explain how changes in environmental conditions can shift the distribution of traits within a population across generations.	• Explain that adaptation results from natural selection acting on inherited variation within a population. (B-LS4-4) • Explain that adapted populations are characterized by traits that increase survival and reproduction (fitness) in a specific environment. (B-LS4-4) <i>Extended Knowledge</i> • Identify that adaptation improves average population success in a given environment but does not guarantee survival of all individuals. (B-LS4-5)	• Explain adaptation as a population-level outcome resulting from differential survival and reproduction under specific environmental conditions. • Explain how changing or conflicting selective pressures can influence adaptive outcomes in populations. • Explain why adaptation does not guarantee population persistence when environmental change is rapid or severe.

B-LS4-4. Construct an explanation based on evidence for how natural selection leads to adaptation of populations.

LS4.C: Adaptation

Eighth Grade	Biology 1	Biology 2
• Explain that adaptation occurs over many generations when natural selection favors traits that improve survival and reproduction under specific environmental conditions (fitness). • Describe adaptation as a population-level process rather than a change within an individual organism's lifetime. • Explain that traits supporting successful survival and reproduction tend to become more common in a population over time. • Explain how changes in environmental conditions can shift the distribution of traits within a population across generations.	• Explain that adaptation results from natural selection acting on inherited variation within a population. (B-LS4-2, • Explain that adapted populations are characterized by traits that increase survival and reproduction (fitness) in a specific environment. (B-LS4-2 • Describe adaptation as a population-level outcome that emerges over many generations rather than a change within individuals. (B-LS4-5) • Describe adaptation, extinction, and divergence as alternative long-term outcomes of population responses to environmental change. (B-LS4-5)	• Explain adaptation as a population-level outcome resulting from differential survival and reproduction under specific environmental conditions. • Explain how changing or conflicting selective pressures can influence adaptive outcomes in populations. • Explain why adaptation does not guarantee population persistence when environmental change is rapid or severe.

B-LS4-5. Evaluate the evidence supporting claims that changes in environmental conditions may result in (1) increases in the number of individuals of some species, (2) the emergence of new species over time, and (3) the extinction of other species.

LS4.C: Adaptation

Eighth Grade	Biology 1	Biology 2
- Explain that adaptation occurs over many generations when natural selection favors traits that improve survival and reproduction under specific environmental conditions (fitness). - Describe adaptation as a population-level process rather than a change within an individual organism's lifetime. - Explain that traits supporting successful survival and reproduction tend to become more common in a population over time. - Explain how changes in environmental conditions can shift the distribution of traits within a population across generations.	- Explain that adapted populations are characterized by traits that increase survival and reproduction (fitness) in a specific environment. (B-LS4-2, B-LS4-4) - Describe adaptation as a population-level outcome that emerges over many generations rather than a change within individuals. (B-LS4-4) - Explain how environmental change can affect population success depending on which traits are present. - Describe adaptation, extinction, and divergence as alternative long-term outcomes of population responses to environmental change. (B-LS4-4) <i>Extended Knowledge</i> - Identify that adaptation improves average population success in a given environment but does not guarantee survival of all individuals. (B-LS4-2)	- Explain adaptation as a population-level outcome resulting from differential survival and reproduction under specific environmental conditions. - Explain how changing or conflicting selective pressures can influence adaptive outcomes in populations. - Explain why adaptation does not guarantee population persistence when environmental change is rapid or severe.